# Jamie Fulford

jamiehf2@illinois.edu jfulfo.github.io linkedin.com/in/jamie-fulford github.com/jfulfo

## Education

**University of Illinois Urbana-Champaign** **August 2025 – Present**
*Ph.D. Computer Science — Advised by Talia Ringer* *Champaign, Illinois*

- **Research:** proof autoformalization, programming languages, formal methods, interactive theorem proving

**University of Virginia** **August 2022 – May 2025**
*B.S. Computer Science and Mathematics* *Charlottesville, Virginia*

- **GPA:** 3.9

## Publications

**BeePL: Correct-by-compilation kernel extensions** — *arXiv preprint* arXiv:2507.09883 **2025**

- Swarn Priya, Frédéric Besson, Connor Sughrue, Tim Steenvoorden, **Jamie Fulford**, Freek Verbeek, Binoy Ravindran

## Research Experience

**University of Illinois Urbana-Champaign** **August 2025 – Present**
*Graduate Research Assistant* *Champaign, Illinois*

- **DARPA expMath:** Python tooling for **proof autoformalization** and **proof decomposition** for **Agda**, targeting homotopy type theory — decomposing a theorem statement into multiple lemmas dispatched to **LLM-based provers**
- **Proof translation between program logics:** designing languages with specific interpretable properties, then implementing them in **Rocq** to interoperate and compile proofs between the two

**Virginia Tech Systems Software Research Group** **January 2025 – July 2025**
*Research Assistant (Proof Engineer)* *Remote*

- Developed type systems and compiler verification for **Linux eBPF** in **Rocq**, proving kernel safety properties
- Extended the verified compiler framework **CompCert** to a new frontend language

**University of Pennsylvania** **May 2024 – August 2024**
*Research Fellow, NSF REU (advised by Steve Zdancewic)* *Philadelphia, Pennsylvania*

- Designed and formalized a novel lambda calculus for parallel and distributed computation in **Rocq**, with machine-checked proofs of **type safety and confluence**

## Software & Formalizations

**Formalizing LLM Guardrails with AgentSpec in Lean 4** Paper Code **Fall 2025**

- Sole author. Executable Lean 4 semantics for **AgentSpec**, a DSL for runtime enforcement of LLM agent behavior, parameterized over predicate tables and external effects, so the guarantees hold for any user-defined predicates
- 42 machine-checked theorems, no `sorry`: **safety is compositional in both directions**, the safety evaluator correctly implements the violation semantics (no false positives or misattribution), per-enforcement contracts, and termination

**ProbaLean: A Probabilistic DSL in Lean 4** Paper Code **Spring 2026**

- Joint, three authors (equal contribution). Discrete probabilistic language in Lean 4 with **denotational semantics over s-finite measures** on Mathlib, plus executable operational semantics
- Proved the interpretation is a **monad morphism** into the Giry monad, **commutativity of independent sampling**, and **soundness** of denotational against operational semantics — axiom-free at $\mathbb{Q}_{\geq 0}$

## Teaching

**Head TA**, Computer Systems & Organization, UVA (500+ students) **Fall 2023, 2024**

## Awards

**PLMW Scholarship** — Programming Languages Mentoring Workshop, POPL 2025 **January 2025**

## Technical Skills

**Proof Assistants**: Agda, Rocq (Coq), Lean 4
**Verification**: Iris, CompCert, Mathlib
**Languages**: OCaml, Haskell, Rust, Python, C, C++, Java, JavaScript, SQL, Bash
**Tools**: Git, Linux, NixOS, LaTeX, PyTorch